BikeSim Vehicle

Overview of BikeSim Vehicle Math Model 1
Bodies and degrees of freedom 1
Aerodynamics 6
Overview of Vehicle Screens 7
Vehicle Assembly 7
Vehicle: Sprung Mass 12
Suspension Kinematics, Zero Jounce, and Wheel Center Heights 12
Coordinate System 13
User Settings 13
Vehicle: Rider Body Mass 16
Preloading and Initial Suspension Deflection 16
Coordinate Systems and Body Movements 16
User Settings 18
References 21

Overview of BikeSim Vehicle Math Model

The BikeSim math model is based on a core model developed and validated by Professors Robin Sharp, *et al*. [1, 2] at Imperial College, London using AutoSim [3, 4] (AutoSim is an earlier version of the multibody symbolic simulation code generator VehicleSim Lisp) from Mechanical Simulation. The math model has been extended to include a full powertrain, full nonlinear kinematics suspension, an enhanced rider positioning control, several enhanced tire model options, additional degrees-of-freedom, and more extensions needed to provide an accurate simulation of most existing motorcycle design concepts.

Bodies and degrees of freedom

Figure 1 and Figure 2 show the geometry of the BikeSim vehicle math model in a side view and a longitudinal view, respectively. The figures also indicate the major rigid bodies contained in the model. Table 1 and Table 2 summarize the numbers of the rigid bodies and degrees of freedom.

The rigid bodies are defined using a parent/child concept. A “child body” is specified relative to a “parent,” and its motion is defined relative to the parent. Details for each body are described in the following subsections.

Note: This document mainly describes the features of the two-wheeled motorcycle models. In version 9.0, a three-wheeled motorcycle model is added and the additional features for the three-wheeled model are described in a separate Help document *Three-wheeled motorcycles*.

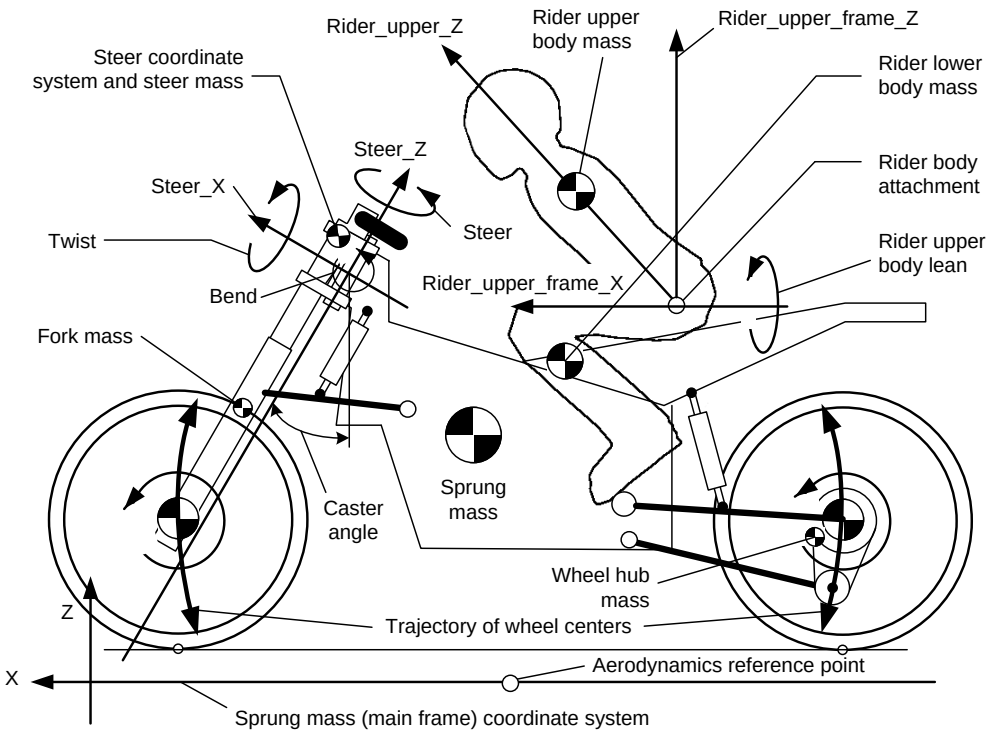


Figure 1. Schematic of the BikeSim vehicle model (side view)

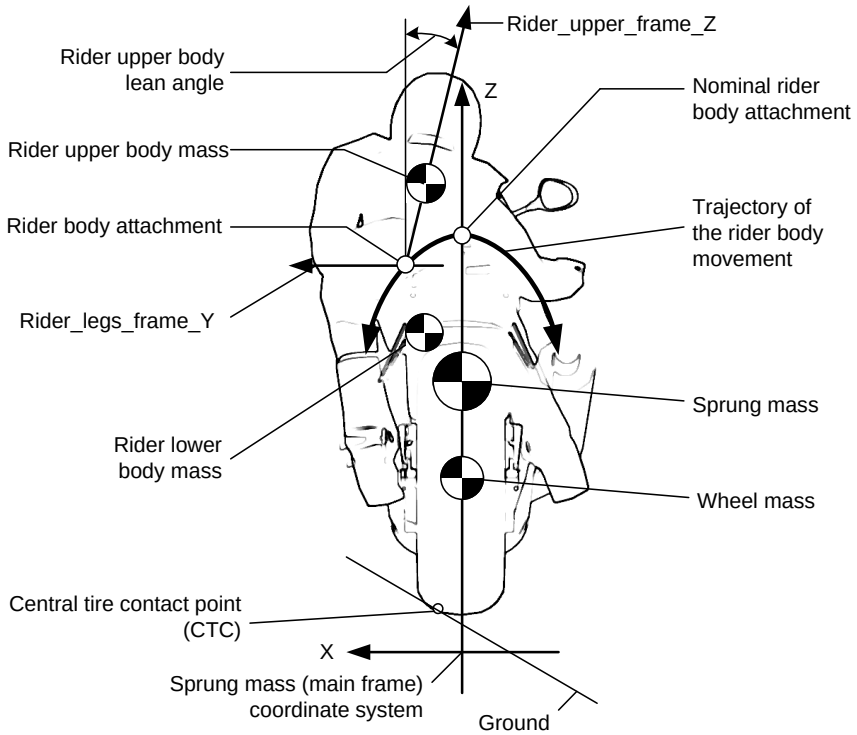


Figure 2. Schematic of the BikeSim vehicle model (longitudinal view from rear)

Table 1. Summary of rigid bodies involved in BikeSim vehicle math model that have both mass and inertia.

Bodies	**Number of bodies**
Main frame	1
Steering head	1
Front fork	1
Rear wheel hub	1
Spinning wheels	2
Rider upper body	1
Rider lower body	1
Engine crank shaft	1
Gear box sprocket (or driveshaft)	1
Total	**10**

Table 2. Summary of degrees of freedom (DOF) involved in the BikeSim vehicle math model.

Bodies	**Number of DOF**
Main frame translation (X,Y,Z)	3
Main frame rotation (X,Y,Z)	3
Steer (steer, twist and bend)	3
Front fork stroke	1
Rear wheel hub (stroke, bend and twist)	3
Wheel spin	2
Rider upper body lean and lateral translation	2
Tire delayed forces (longitudinal and lateral)	4
Brake fluid pressure	2
Engine crank shaft	1
Engine throttle delay	1
Gear box sprocket (or driveshaft)	1
Fuel consumption	1
Suspension spring friction	2
“Locked wheel” equations	2
Total	**32**

Main frame

1. The **main frame (Sprung Mass)** includes all mechanical components above the front and rear suspensions (body frame, fuel tank, engine, transmission, seat, rear fender, etc.) except

steering parts. The local coordinate system origin is defined by the user parameters. This main frame body involves six degrees of freedom (DOF).

Steering system

2. The **twist and bending massless frame** is a massless frame that represents the twist and bending motion of the steering system. The body coordinate system has its origin at the point of attachment of the steering head to the main frame. The vertical axis is aligned with the nominal steer axis. This body has two DOF: twist (Steer_twist_X) and bend (Steer_twist_Y).

3. The **steering head** is the sprung mass of the front suspension. It includes the masses of all components that move with steering minus the unsprung masses associated with the wheel, brake, and moving suspension and body parts. The body is attached to the twist and bending massless frame at the steering head attachment. Therefore, if the massless frame rotates relative to the main frame in the twist and bending directions, the steering head also rotates together. Furthermore, the steering head involves the local coordinate system and it involves one degree of freedom of the steer rotation about the steer axis (Steer_Z).

Note: Starting with version 2018.1, the twist and bending flexibility can be enabled or disabled by a flexible-frame option. The detail setting is described in the **Vehicle Assembly** section.

Suspension systems

4. The **front fork** is the unsprung mass of the front suspension. This includes all front unsprung components that do not spin with the wheel. For example, the lower forks, fender, and brake caliper are included while the wheel, tire, and brake rotor are not.

   Most motorcycle front suspensions involve a conventional structure in which the steer and cushion mechanism are in one unit and the both steering head and front wheel rotate together about the steer axis (e.g. telescopic front fork suspension). However, there are alternatives to the conventional front fork, such as *hub center steer system* (Bimota Tesi), *double wishbone* (BMW Duolever, Yamaha GTS), and *McPherson strut* (BMW Telelever). In those systems, the steering head and wheel are separated by employing various types of linkage systems, and the steer axis for the wheel varies with the suspension stroke. More details are available in BikeSim **Help** documents for the steering system (*BikeSim Steering System*) and suspension system (*BikeSim Suspension Systems*). BikeSim supports two cases: one for the fixed steer axis relative to main frame and the other for variable steer axis with wheel dive. The multibody definition of the front fork body is different in those cases:

   a) **Fixed steer axis**. The parent is the steering head body and the fork steers together with the steering head about Steer_Z. The body involves three motions relative to the parent: longitudinal translation Steer_X, vertical translation Steer_Z (suspension stroke), and rotation angle Fork_Y (dive of the fork or wheel hub).

However, two of those motions (Steer_X and Fork_Y) are kinematical constrained as nonlinear functions of vertical motion.

b) **Variable steer axis.** The parent is the twist and bending massless frame. The body has four motions relative to the parent: the longitudinal translation Steer_twist_X, vertical translation Steer_twist_Z (suspension stroke), rotation angle Fork_Y (dive of the wheel hub), and rotation angle Fork_Z (steer of the wheel hub). The order of the rotations is: first dive, then steer. Therefore, the steer axis can change with the dive rotation. However, two of the motions (longitudinal translation and dive rotation) are constrained as nonlinear functions of vertical motion. Also, the steer rotation is constrained with respect to the steer head angle; although those steer axes are different, the steer angles are the same.

Overall in the both cases, the fork has only one independent DOF: suspension stroke.

5. The **rear wheel hub (swing arm)** is the unsprung mass of the rear suspension. This mass includes the all rear unsprung components that do not spin with the wheel. For example, the wheel hub, brake caliper, drive gear housing (if shaft drive), and chain sprocket (if chain drive) are included while the wheel, tire, and brake rotor are not. If the rear suspension is a single swing arm with a single pivot, it also includes the swing arm mass. In this case, the wheel hub and swing arm are treated as one unit. However, when simulating more complicated suspension structures, e.g. parallel linkage with torque reaction rods, the unsprung mass excludes the swing arm mass because the wheel hub and swing arm have different angles and the unsprung mass represents the wheel hub. The rear wheel hub body is attached to the main frame at the rear wheel axle location. The body involves four motions relative to the main frame: the longitudinal translation in X, vertical translation in Z (suspension stroke), lateral translation in Y (lateral bend) and rotation angle Hub_Y (dive of the hub or swing arm). Two of those motions (longitudinal translate and dive rotation) are constrained with respect to the vertical motion.

   Furthermore, two additional motions (steer in Hub_Z and twist in Hub_X in sequential order) are added from the above four motions. And the steer in Hub_Z is constrained with respect to the lateral translation. Therefore, the rear wheel hub has total three independent DOFs: suspension stroke, lateral displacement (bend) and twist.

Note: Starting with version 2020.1, the bending and twist flexibility of rear swing arm can be enabled or disabled by a flexible-frame option. The detail setting is described in the separate document, *BikeSim Suspension System*.

Wheels

6. The **front and rear spinning wheels.** Each of these masses includes all components that spin with the wheel. For example, the wheel, tire, and brake rotor. The spinning wheels are attached on the front fork and rear wheel hub at the wheel axle locations. Each body involves one DOF for wheel spin.

Rider

7. The **rider lower body** includes legs and feet. The upper and lower body sections are divided approximately at the rider's hips. NASA suggests that about 38% of the total mass of the human body is associated with the lower body. The lower body is attached on the main frame at seat location (rider body attachment). However, strictly speaking, the lower body involves two separated local coordinate systems: one is an intermediate massless frame (Rider_legs_frame), which represents the motion of the lower body, and the other is an actual mass. The massless frame involves two motions relative to the main frame: lateral translation in Y and vertical translation in Z. The vertical translation constrained with respect to the lateral translation. The rider upper body (described later) is attached on this massless frame at the hip point. On the other hand, the mass body involves two translate motions (in Y and Z) relative to the main frame. They are also constrained with respect to the lateral translation of the massless frame. Therefore, overall, the rider lower body has only one DOF.

8. The **rider upper body** includes torso, head, arms, etc. The upper and lower body sections are divided approximately at the rider's hips. NASA suggests that about 62% of the total mass of the human body is associated with the upper body. The upper body is attached on the lower body frame at hip position through an intermediate massless frame (Rider_upper_frame). The rider upper body's forward lean angle is fixed by a setting parameter. The body involves a lateral leaning motion about Rider_upper_frame_X relative to the main frame and its leaning angle is defined by the angle between Rider_upper_frame_Z and vertical axis of the main frame (Z).

Powertrain

9. The **engine crankshaft** includes the spin inertia but does not involve the mass. The crankshaft is used to calculate the engine speed as response to the engine torque applied to the crankshaft and torque reaction from the torque transfer devise (e.g. mechanical clutch or hydraulic torque converter). Also, the engine crankshaft body includes a half of the torque transfer device (engine side).

10. The **gearbox sprocket or driveshaft** rigid body definition depends on the drive system (chain drive or shaft drive system). If a chain drive system is selected, the body represents the gearbox sprocket, while the body represents the driveshaft for a shaft drive system. The body does not involve a mass. However, the body includes all spin inertia properties, associated with the gearbox sprocket or driveshaft: half of the torque transfer device (transmission side), transmission, gearbox, etc.

Aerodynamics

BikeSim includes full aerodynamics (three forces and three moments) and more details are available in BikeSim **Help** documents for the aerodynamics (*Aerodynamics*). However, the previous version (until version 2.0) includes only the longitudinal characteristics, and such simplified option is still available in **Vehicle: Assembly** screen. In the simplified option, the drag (F_{X_Aero}), lift (F_{Z_Aero}) and pitching moment (M_{Y_Aero}) that both act on a point on the sprung mass called the aerodynamic reference point, which is at ½ the wheelbase, at the level of X axis of the

sprung mass coordinate system (see Figure 1). If changes are made to the wheelbase, the location of the aerodynamic reference point is automatically changed. The moment is applied at the sprung mass body. The aerodynamics forces and moment are calculated by the equations:

$$\begin{aligned} F_{X_Aero} &= -0.5 \times D_{Aero} \times C_D \times A_{Aero} \times V_X{}^2 \\ F_{Z_Aero} &= 0.5 \times D_{Aero} \times C_L \times A_{Aero} \times V_X{}^2 \\ M_{Y_Aero} &= -0.5 \times D_{Aero} \times C_P \times A_{Aero} \times L_{WB} \times V_X{}^2 \end{aligned} \tag{1}$$

where D_{Aero} is air density, C_D is drag coefficient, C_L is lift coefficient, C_P is pitch moment coefficient, A_{Aero} is aerodynamic cross section and V_X is vehicle forward speed. The forces and moment are based on the ISO coordinate system. Therefore, according to Equations (1), the positive drag coefficient causes negative longitudinal force (resisting forward movement), while the positive lift coefficient causes positive vertical force (lifting body). The positive pitch moment coefficient causes negative pitch moment (front nose up).

The coefficients for the above equations are set on the **Vehicle: Assembly** screen. The air density has a default value of 1.206 kg/m^3; alternate values can be set using the full aerodynamics screen (*Aerodynamics*).

Overview of Vehicle Screens

This document describes the following screens.

1. The **Vehicle: Assembly** screen is associated with the main vehicle link on the **Run Control** screen. This screen assembles the vehicle from datasets that define the major systems and components, and aerodynamics. The vehicle as defined by this screen is in the design load condition (sometimes called curb condition).
2. The **Vehicle: Sprung Mass** screen is linked from the **Vehicle: Assembly** screen. This screen sets size and weight information about the sprung mass of the main vehicle.
3. The **Vehicle: Rider Body Mass** screen is linked from the **Vehicle: Assembly** screen. This screen sets size and weight information about the rider upper and lower bodies.

Vehicle Assembly

Figure 3 shows the **Vehicle: Assembly** screen on which the components and systems of a motorcycle or scooter is assembled.

① Link to a **Vehicle: Sprung Mass** dataset that defines the properties of the sprung mass.

② Link to a **Vehicle: Rider Body Mass** dataset that defines the mass properties and position of the rider.

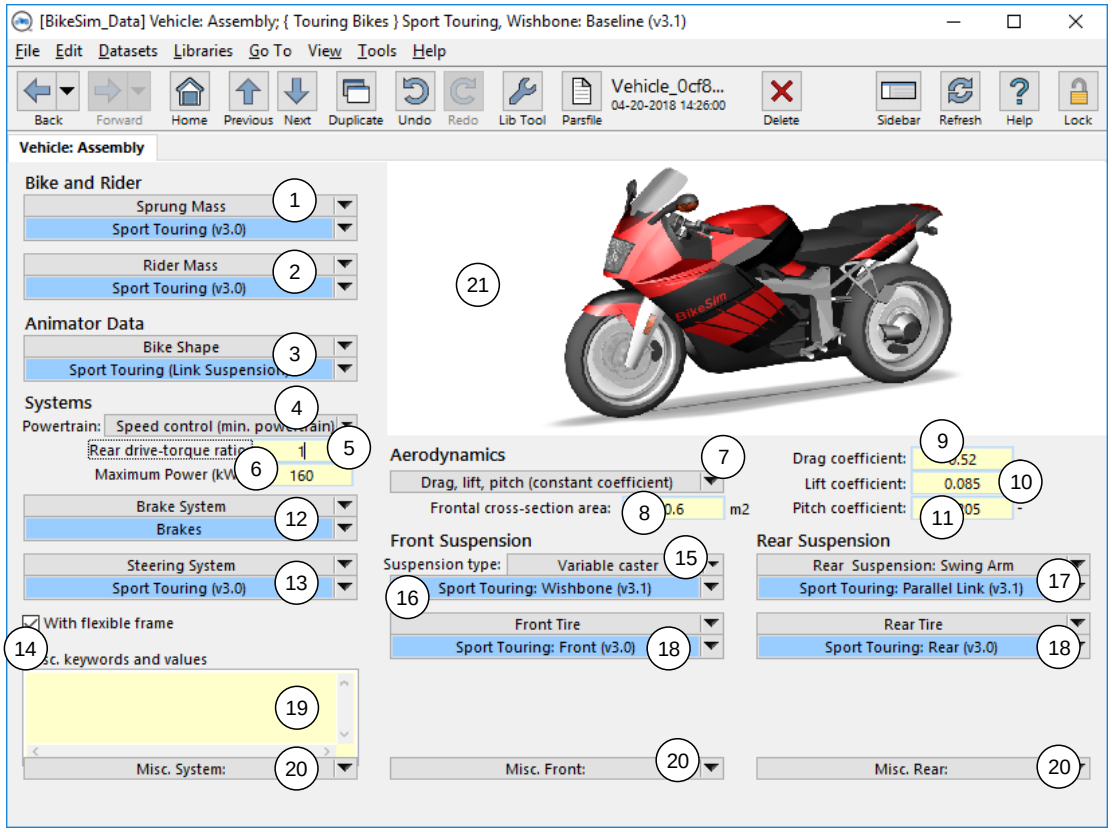
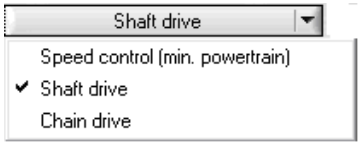


Figure 3. Vehicle: Assembly screen.

③ Link to an animator dataset. This is usually a dataset from the library **Animator: Bike Shape Assembly**, but can be any dataset that defines the shape of the sprung mass for the animator (e.g. **Animator: Vehicles and Sensor Targets** for car shape). If the linked dataset has an associated image, that image is automatically shown on this screen ⑳ when the link is made.

④ Drop-down list for specifying the type of powertrain. The option chosen determine the kind of control that appears directly underneath.



⑤ Driveline information associated with type selected with the above control ④.

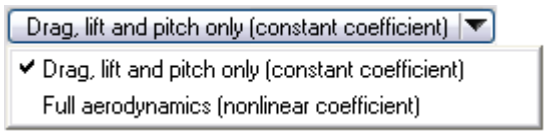
1. If the powertrain is **Speed control (min. powertrain)** as shown in Figure 3, a data field is shown with the ratio of drive torque (from the speed controller) that goes to the rear wheels (keyword = `R_REAR_DRIVE_SC`). Use 1 for simple rear-wheel drive. A small number of motorcycles today have two-wheel drive. Two-wheel-drive can be simulated by specifying a value for the rear drive torque ratio between zero and one (e.g., 0.5 is a 50/50 torque split.)

2. If the powertrain is **Shaft drive**, a link is shown to a **Powertrain: Shaft Drive** dataset.

3. If the powertrain is **Chain drive**, a link is shown to a **Powertrain: Chain Drive** dataset.

⑥ If the powertrain ④ is **Speed control (min. powertrain**) as shown in Figure 3, a data field is shown that limits the power used by the speed controller (keyword = `PMAX_SC`). The speed controller uses proportional-integral control to try to match a target speed. The power being transferred to the wheels is monitored and not allowed to exceed this limit.

⑦ Drop-down list for specifying the type of aerodynamics. The option chosen determine the kind of control that appears directly underneath.



1. If the aerodynamics is **Drag, lift and pitch only (constant coefficient)** as shown in Figure 3, data fields are shown with the constant coefficients and cross-section area (⑧ - ⑪).

2. If the aerodynamics is **Full aerodynamics (nonlinear coefficient)**, a link is shown to a **Aerodynamics: Main Screen** dataset.

⑧ Aerodynamic cross section area (keyword = `AREA_AERO`). This area appears in the equations that determine the aerodynamic forces and moments presented for drag, lift, and pitch on page 6 and repeated below. BikeSim calculates aerodynamic effects assuming still air with a density of 1.206 kg/m^3; alternate values can be set using the full aerodynamics model on the **Aerodynamics: Main Screen**.

⑨ Aerodynamic drag coefficient (keyword = `C_AERO_D`). The drag force is calculated by:

$$F_{X_Aero} = -0.5 \times D_{Aero} \times C_D \times A_{Aero} \times V_X^{\ 2}$$

where F_{X_Aero} is drag force, D_{Aero} is air density, C_D is drag coefficient, A_{Aero} is aerodynamic cross section and V_X is vehicle forward speed. The drag force is applied at a reference point at ½ the wheelbase, at the level of X axis of the sprung mass (main frame) coordinate system, in the direction of X axis. The positive drag coefficient causes negative longitudinal force (resisting forward movement).

⑩ Aerodynamic lift coefficient (keyword = `C_AERO_L`). The lift force is calculated by:

$$F_{Z_Aero} = 0.5 \times D_{Aero} \times C_L \times A_{Aero} \times V_X^{\ 2}$$

where F_{Z_Aero} is lift force, D_{Aero} is air density, C_L is lift coefficient, A_{Aero} is aerodynamic cross section and V_X is vehicle forward speed. The lift force is applied at a reference point at ½ the wheelbase, at the level of X axis of the sprung mass (main frame) coordinate system, in the direction of Z axis. The positive lift coefficient causes positive vertical force (lifting body).

⑪ Aerodynamic pitch moment coefficient (keyword = `C_AERO_P`). The pitch moment is calculated by:

$$M_{Y_Aero} = -0.5 \times D_{Aero} \times C_P \times A_{Aero} \times L_{WB} \times V_X^{\ 2}$$

where M_{Y_Aero} is pitch moment, D_{Aero} is air density, C_P is pitch moment coefficient, A_{Aero} is aerodynamic cross section, L_{WB} is wheelbase and V_X is vehicle forward speed. The pitch moment is applied at the sprung mass (main frame) body. The positive pitch moment coefficient causes negative pitch moment (front nose up).

⑫ Link to a **Brake System** dataset.

⑬ Link to a **Steering System** dataset.

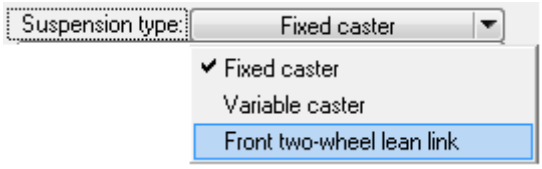
⑭ Checkbox to enable twist and bending flexibility of the front fork frame. The characteristics for twist and bending flexibility are set on three different libraries: **Steering System**, **Suspension: Fork**, **Suspension: Lean Link System**. If this is unchecked, the front fork frame is treated as rigid, the fork doesn't twist and bend relative to the main frame, and all parameters in Table 3 are ignored.

> **Note:** In the previous versions, the front fork was always flexible for twist and bend. Therefore, the box will be checked as default when the vehicle data is imported from previous versions.

Table 3. Summary of setting parameters for twist and bending flexibility of the front fork frame.

Description	**Keyword**	**Library (screen)**
Frame-twist stiffness	`K_FRAME_TWIST`	Steering System
Frame-twist damping	`D_FRAME_TWIST`	
Fork longitudinal stiffness	`F_AXLE_X`	Suspension: Fork
Fork longitudinal damping	`F_AXLE_VX`	
Frame-twist stiffness	`K_LEAN_LINK_TWIST`	Suspension: Lean Link System
Frame-twist damping	`D_LEAN_LINK_TWIST`	

⑮ Drop-down list for specifying the 3D multibody interaction between the steering and the front suspension kinematics.



1. **Fixed caster:** In this case, the steering head and front wheel rotate together about a steer axis (Steer_Z) whose orientation does not change with respect to the main frame.

2. **Variable caster:** In this case, the steer axis for the front wheel may vary with suspension stroke. The steer axis for the steering head is still fixed on the main frame. However, the steer axis of the front wheel varies with the wheel hub's dive angle, which is specified on **Suspension: Dive Angle (Caster Change)** screen. The details of those settings are described in the **Help** document *Suspension System*.

3. Front two-wheel lean link: In this case, another wheel is added on the front-right side and the vehicle becomes a three-wheeled motorcycle. In this case, an additional link for a third tire also appears on the screen. The front suspension blue field links to the **Suspension: Lean Link System** library. The details of those settings are described in the **Help** document under the **Vehicles** category, *Three-wheeled motorcycles*.

Note The above selection control ⑮, checkbox ⑭, and rear swing arm link ⑰ are used by the browser to configure between twelve different combinations of model layout: `fixed_caster`, `variable_caster`, and `three_wheeled` with and without prefix `rigid_` and suffix `_flex`. The model layout code appears above the **Vehicle** blue link on the **Run Control** screen (see Table 4 for summary).

The detail setting for rear swing arm flexibility is described in the separate document, *BikeSim Suspension System*.

Table 4. Summary of model layout codes for the front and rear suspension types.

Suspension types ⑮	**Suspension flexibility**		**Model layout code**
	Front fork ⑭	**Rear swing arm** ⑰	
Fixed caster	Checked	`Swing Arm`	`fixed_caster`
	Not checked	`Swing Arm`	`rigid_fixed_caster`
	Checked	`Flex Swing Arm`	`fixed_caster_flex`
	Not checked	`Flex Swing Arm`	`rigid_fixed_caster_flex`
Variable caster	Checked	`Swing Arm`	`variable_caster`
	Not checked	`Swing Arm`	`rigid_variable_caster`
	Checked	`Flex Swing Arm`	`variable_caster_flex`
	Not checked	`Flex Swing Arm`	`rigid_variable_caster_flex`
Front two-wheel lean link	Checked	`Swing Arm`	`three_wheeled`
	Not checked	`Swing Arm`	`rigid_three_wheeled`
	Checked	`Flex Swing Arm`	`three_wheeled_flex`
	Not checked	`Flex Swing Arm`	`rigid_three_wheeled_flex`

⑯ Link to a **Suspension: Fork** dataset for a two-wheeled vehicle or a **Suspension: Lean Link System** dataset for a three-wheeled vehicle.

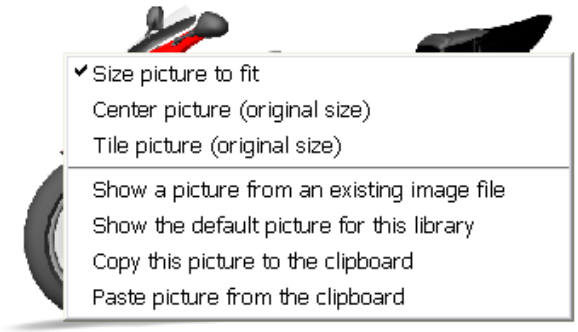
⑰ Link to a **Suspension: Swing Arm** dataset.

⑱ Links to **Tire** datasets for the two tires.

⑲ Miscellaneous field. This is a location for specifying parameters for extensions to the model or information for the animator.

⑳ Miscellaneous links. These have no predefined purpose, but can be linked to datasets that specify parameters for extensions to the model or information for the animator. The data from the first link applies to the sprung mass or whole vehicle. Data from the other two links apply to the front or rear suspension, as indicated by their locations on the screen.

㉑ Image of vehicle. When you change the link to the animator dataset ③, the associate image from that dataset is shown here. You can also specify an alternate image by clicking on the image to display a drop-down list of options. See the *VS Browser Reference Manual* for details about these options.



Vehicle: Sprung Mass

The parameters and summary information for the sprung mass are listed in a section of the Echo file as shown in Figure 4. The **Vehicle: Sprung Mass** screen is used to define the main size and weight properties of a motorbike without rider or extra loads.

This configuration defined on this screen is called the design condition in BikeSim, and in other applications is sometimes called the curb condition. Besides defining some of the main mass and inertia properties of the vehicle, the properties on this screen affect the interpretation of data from other screens that depend on load.

In addition to defining the design load condition, the sprung mass defines the locations of points in the vehicle model with its coordinate system.

Suspension Kinematics, Zero Jounce, and Wheel Center Heights

Suspension motions are tied to vertical movement called jounce that provides the reference for many forms of nonlinear suspension kinematical data. These include longitudinal movement, suspension dive (caster change), and suspension lever ratio. In particular, the definition of zero jounce is critical for properly using this data.

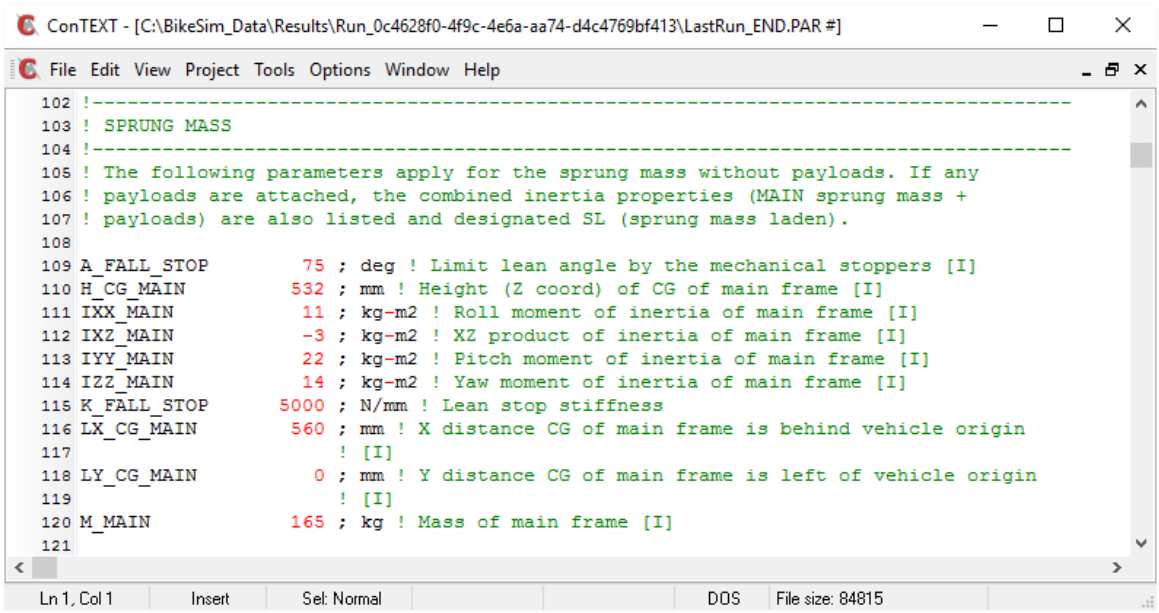


Figure 4. Sprung mass parameters in the Echo file.

This screen shows the location of the wheel centers when the front and rear springs are in equilibrium supporting the sprung mass in a level condition, where roll and pitch angles are zero. The heights of the wheel centers in a specific run will depend on the road or ground geometry, the tire sizes and properties, and the presence of additional loads (i.e., rider mass and payloads). However, this screen contains all the information needed to define the concepts of zero pitch and roll and suspension jounce.

Coordinate System

Loads, sensors, and various reference points are specified using X, Y, and Z coordinates in a coordinate system fixed in the sprung mass. The axis directions and the location of the origin of the coordinate system are defined specifically for the vehicle in the position shown on this screen. In BikeSim, the origin of the sprung mass coordinate system is at the front wheel center in the longitudinal position. The height position of the origin can be defined by users specifying the height of front and rear wheel center and center mass. The height position of the origin can be ground level, wheel center or wherever user wants to define. However, once define the origin of the coordinate system the definition should be consistent among other screens, which use the sprung mass coordinate system for the data settings.

User Settings

Figure 5 shows the **Vehicle: Sprung Mass** screen with the controls described below.

① Animator reference height (keyword = `Z_LENGTH`). The animator resizes the shape information to match vehicle dimensions. This dimension is used to stretch or shrink the vertical scale of the shape used for the vehicle sprung mass.

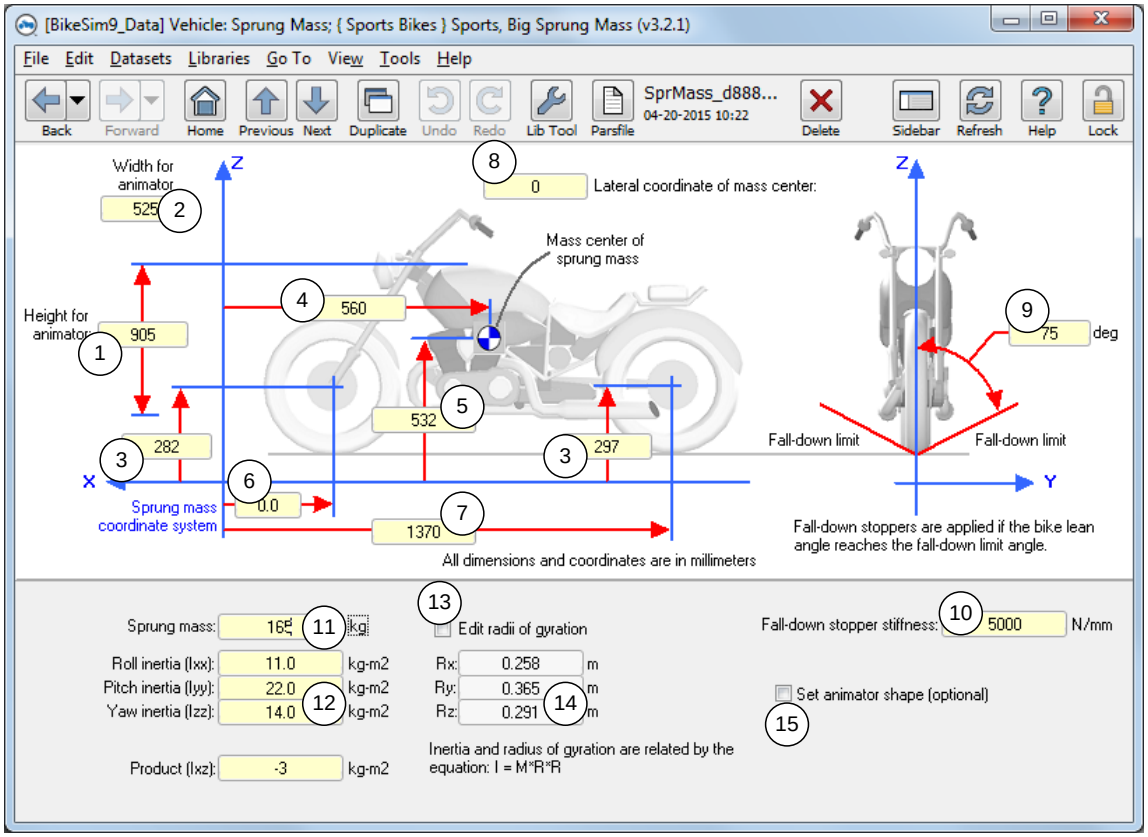


Figure 5. Vehicle: Sprung Mass screen.

② Animator reference width (keyword = `Y_LENGTH`). The animator resizes the shape information to match vehicle dimensions. This dimension is used to stretch or shrink the width of the shape used for the vehicle sprung mass.

③ Heights of the wheel spin axes above the sprung mass origin at the design load condition with zero pitch and roll (database keywords = `*HWC_F`, `*HWC_F`; solver keyword = `H_WC (IAXLE)`). These define the relationship between the locations of the wheels relative to the sprung mass. They also determine how the sprung mass is located relative to the wheels in the animator.

④ Distance that the mass center (design condition) lies behind the sprung mass origin (Keyword = `LX_CG_MAIN`). This parameter and the wheelbase determine how much of the mass is supported by the front and rear suspensions in the design condition.

⑤ Height of mass center above the origin in design condition (keyword = `H_CG_MAIN`).

⑥ Distance that the front wheel center (design condition) lies behind the sprung mass origin (Keyword = `LX_AXLE(1)`).

⑦ Distance that the rear wheel center (design condition) lies behind the sprung mass origin (Keyword = `LX_AXLE(2)`). Wheelbase is the distance between two wheel centers of front and rear that is give by: `LX_AXLE(2)-LX_AXLE(1)` .

This value is also passed to the animator (keyword = `X_LENGTH`) to size the sprung mass shape longitudinally.

⑧ Lateral coordinate of mass center (keyword = `LY_CG_MAIN`). Normally zero, this can be given a non-zero value if the vehicle is not laterally symmetric. A positive value means the mass center is located on the left side of the vehicle.

⑨ Fall-down limit angle. To enable BikeSim to simulate the case of a bike falling down or the effects of contact between sprung mass and ground, a spring ("fall-down stopper" defined by ⑩) is placed between the sprung mass and ground. This parameter (keyword = `A_FALL_STOP`) describes the lean angle at which this spring becomes effective. It can be used to describe the effect of contact with the ground by foot pegs, exhaust system, or other components.

⑩ Fall-down stopper stiffness (keyword = `K_FALL_STOP`) is the stiffness of a spring between the sprung mass and ground to simulate the case of a bike falling down or the effects of contact between sprung mass and ground. It can be used to describe the effect of contact with the ground by foot pegs, exhaust system, or other components.

⑪ Sprung mass (keyword = `M_MAIN`). This is the mass of the vehicle in the design load configuration minus the unsprung masses associated with the wheels, axles, and moving suspension parts.

⑫ Moments and products of inertia of the sprung mass (design condition). The moments and product are all taken about the mass center of the sprung mass and are based on a vehicle axis system in which the X axis (longitudinal) and Y axis (lateral) are parallel with the ground when the vehicle is at rest on a flat level surface, and the Z axis (vertical) points up in parallel with the gravity vector.

The moments of inertia in roll (`IXX_MAIN`), pitch (`IYY_MAIN`), and yaw (`IZZ_MAIN`) can be entered directly in these fields. Alternatively, they can be calculated using radii of gyration ⑭.

The XZ product of inertia is also about the mass center and is defined as the negative of the volume integral:

$$I_{XZ} = -\int^{V} \rho \, x \, z \, dv$$

The product is positive when the principal X axis of the sprung mass tilts down (looking forward). This value must be entered directly—there are no built-in tools to calculate it.

⑬ **Edit Radii of Gyration** checkbox. Checking this box enables editing of three data fields with X, Y, and Z radii of gyration ⑭, as an alternative to entering moments of inertia. When the box is checked, the moment of inertia fields are locked, but the values displayed in them are updated to reflect changes to the radii of gyration or mass. When it is unchecked, the moment of inertia fields is enabled for editing and the radii of gyration fields are locked, displaying values calculated from the mass and inertias.

⑭ Radii of gyration (not parameters). Sometimes measured values for `Ixx`, `Iyy`, and `Izz` are not available but `Rx`, `Ry`, and `Rz` can be estimated. When the **Edit Radii of Gyration** checkbox is checked, the corresponding moments of inertia are calculated with the equation:

$$I = M*R^2$$

where M is the sprung mass ⑪ and R is the specified radius of gyration.

Use any of these three radii if the corresponding moment of inertia is not available but the corresponding radius can be estimated. If used:

1. a guess for Rx might be a third of the vehicle width,
2. a guess for Ry might be half the wheelbase, and
3. a guess for Rz might be half the wheelbase.

⑮ **(Optional) Animator Shape** checkbox. Check to define the shape of the sprung mass for the animator.

Vehicle: Rider Body Mass

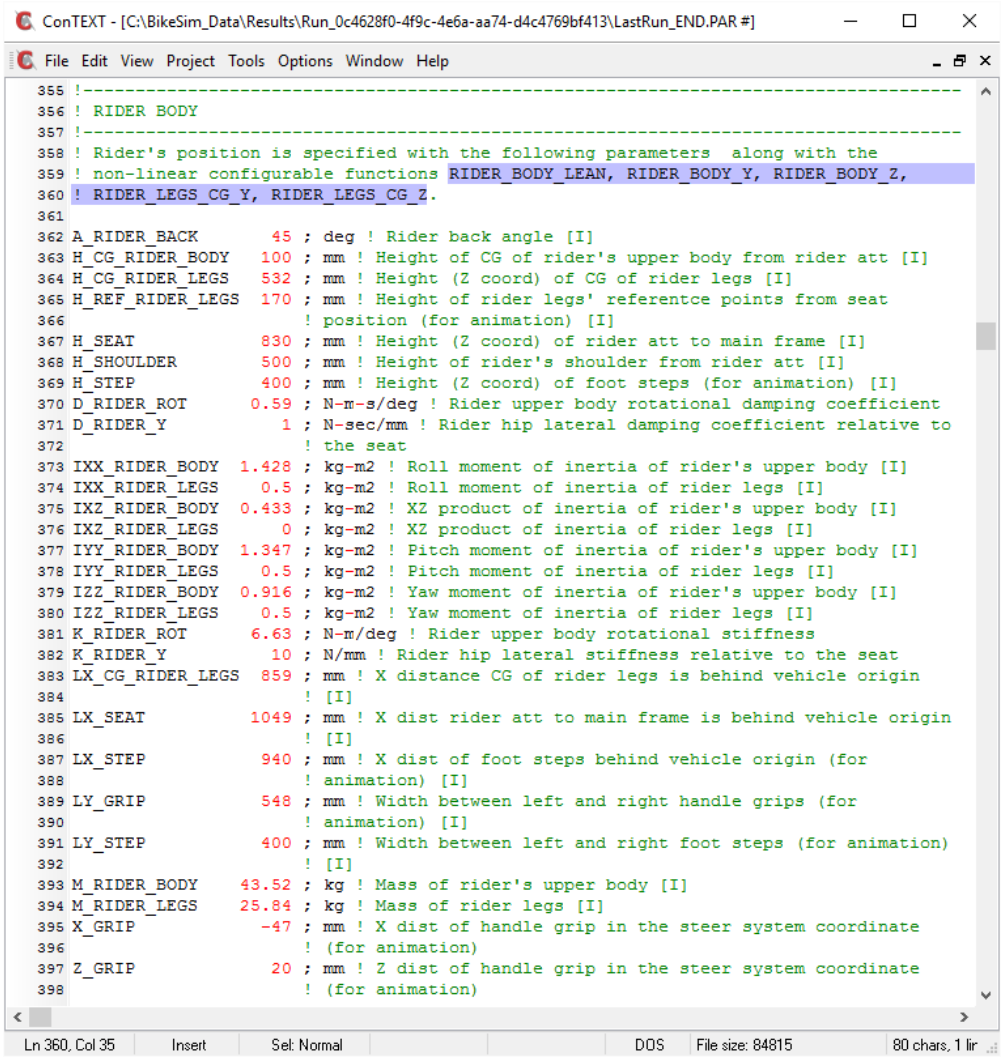
The parameters and summary information for the rider body mass are listed in a section of the Echo file as shown in Figure 6. Gray highlighted configurable functions (lines: 359-360) are used to control riding positions that are described in the separate document, *Rider Controls (Motorcycle)*. The rider of a two-wheeled vehicle can contribute a significant portion of the total vehicle mass and inertia. The **Vehicle: Rider Body Mass** screen (see Figure 7) is used to define the main size and weight properties of the primary rider for the BikeSim models.

Preloading and Initial Suspension Deflection

The initial deflection of the suspensions (the design condition of the motorcycle) is defined in BikeSim without a rider. This means that rider datasets can be changed without the need to revise vehicle suspension parameters. It also means that the dimensions specified on this screen for the rider location depict the vehicle with undeflected suspension. If you take measurements from a physical rider on a motorcycle, the suspension should not be allowed to deflect (for example, place the bike on stands, fixing it in its design position).

Coordinate Systems and Body Movements

To facilitate the study of various riding postures, and to support the ability of the rider to lean relative to the bike, the mass properties of the rider are divided into Upper Body and Lower Body parameters. NASA has suggested a typical distribution of the total rider mass as 62% associated with the upper body and 38% with the lower body.



```
355 !-----------------------------------------------------------------------------
356 ! RIDER BODY
357 !-----------------------------------------------------------------------------
358 ! Rider's position is specified with the following parameters  along with the
359 ! non-linear configurable functions RIDER_BODY_LEAN, RIDER_BODY_Y, RIDER_BODY_Z,
360 ! RIDER_LEGS_CG_Y, RIDER_LEGS_CG_Z.
361
362 A_RIDER_BACK       45 ; deg ! Rider back angle [I]
363 H_CG_RIDER_BODY   100 ; mm ! Height of CG of rider's upper body from rider att [I]
364 H_CG_RIDER_LEGS   532 ; mm ! Height (Z coord) of CG of rider legs [I]
365 H_REF_RIDER_LEGS  170 ; mm ! Height of rider legs' referentce points from seat
366                       ! position (for animation) [I]
367 H_SEAT            830 ; mm ! Height (Z coord) of rider att to main frame [I]
368 H_SHOULDER        500 ; mm ! Height of rider's shoulder from rider att [I]
369 H_STEP            400 ; mm ! Height (Z coord) of foot steps (for animation) [I]
370 D_RIDER_ROT      0.59 ; N-m-s/deg ! Rider upper body rotational damping coefficient
371 D_RIDER_Y           1 ; N-sec/mm ! Rider hip lateral damping coefficient relative to
372                       ! the seat
373 IXX_RIDER_BODY  1.428 ; kg-m2 ! Roll moment of inertia of rider's upper body [I]
374 IXX_RIDER_LEGS    0.5 ; kg-m2 ! Roll moment of inertia of rider legs [I]
375 IXZ_RIDER_BODY  0.433 ; kg-m2 ! XZ product of inertia of rider's upper body [I]
376 IXZ_RIDER_LEGS      0 ; kg-m2 ! XZ product of inertia of rider legs [I]
377 IYY_RIDER_BODY  1.347 ; kg-m2 ! Pitch moment of inertia of rider's upper body [I]
378 IYY_RIDER_LEGS    0.5 ; kg-m2 ! Pitch moment of inertia of rider legs [I]
379 IZZ_RIDER_BODY  0.916 ; kg-m2 ! Yaw moment of inertia of rider's upper body [I]
380 IZZ_RIDER_LEGS    0.5 ; kg-m2 ! Yaw moment of inertia of rider legs [I]
381 K_RIDER_ROT      6.63 ; N-m/deg ! Rider upper body rotational stiffness
382 K_RIDER_Y          10 ; N/mm ! Rider hip lateral stiffness relative to the seat
383 LX_CG_RIDER_LEGS  859 ; mm ! X distance CG of rider legs is behind vehicle origin
384                       ! [I]
385 LX_SEAT          1049 ; mm ! X dist rider att to main frame is behind vehicle origin
386                       ! [I]
387 LX_STEP           940 ; mm ! X dist of foot steps behind vehicle origin (for
388                       ! animation) [I]
389 LY_GRIP           548 ; mm ! Width between left and right handle grips (for
390                       ! animation) [I]
391 LY_STEP           400 ; mm ! Width between left and right foot steps (for animation)
392                       ! [I]
393 M_RIDER_BODY    43.52 ; kg ! Mass of rider's upper body [I]
394 M_RIDER_LEGS    25.84 ; kg ! Mass of rider legs [I]
395 X_GRIP            -47 ; mm ! X dist of handle grip in the steer system coordinate
396                       ! (for animation)
397 Z_GRIP             20 ; mm ! Z dist of handle grip in the steer system coordinate
398                       ! (for animation)
```

Figure 6. Rider body mass parameters in the Echo file.

Upper body

The inertia properties for the upper body are defined about the upper body mass center along a "spinal axis" as if the rider were sitting perfectly erect, parallel to the vehicle coordinates. BikeSim adjusts the upper body mass properties according the forward lean angle specified here and any lateral lean angle. The upper body has a local coordinate system, whose origin is the rider body attachment (see Figure 1), and vertical axis is the spinal axis.

The lateral lean angle may be passive, controlled by a spring and damping value specified here, or active, controlled by information from the **Control: Rider Upper Body Lean Angle vs. Time (Open Loop)** or **Control: Rider Upper Body Lean Angle vs. Station (Open Loop)** screen.

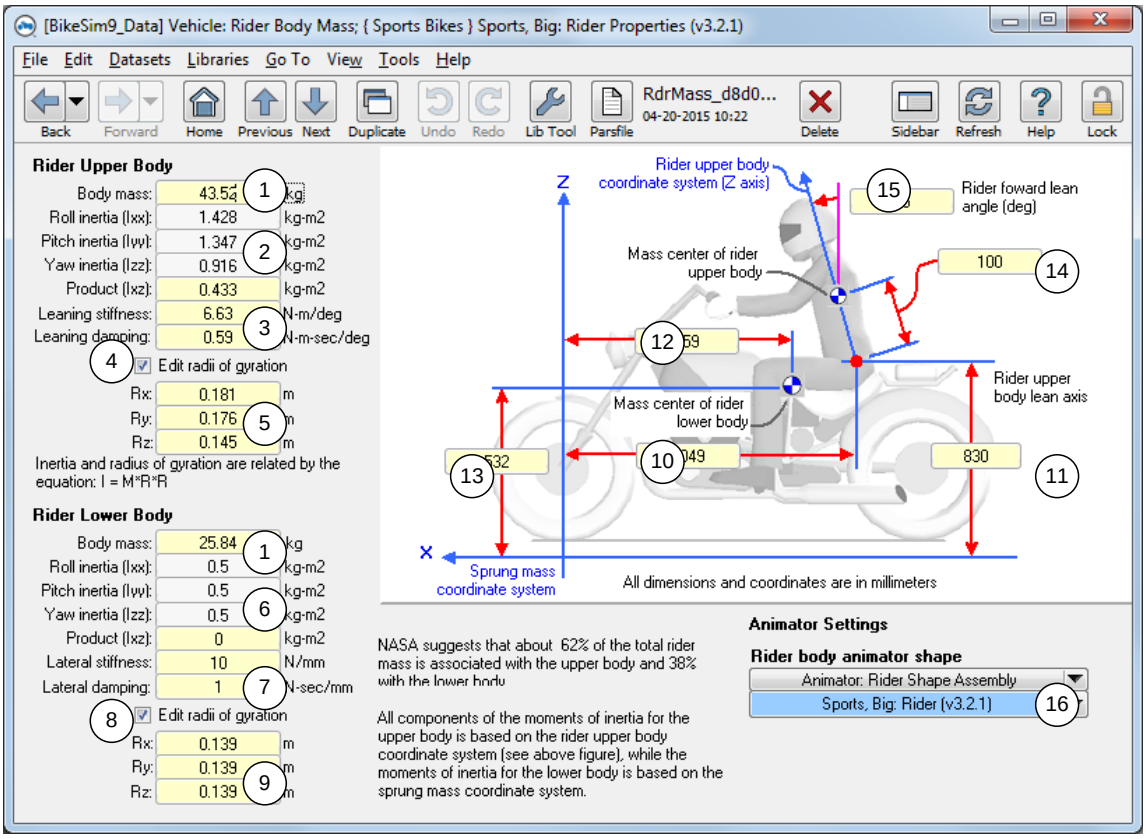


*Figure 7. The **Vehicle: Rider Body Mass** screen.*

Lower body

The lower body has a local coordinate system, whose origin is the rider body attachment, and all axes are parallel to the vehicle sprung mass coordinate system. The inertia properties for the lower body are defined about the lower body CG based on the lower body coordinate system.

The lower body moves laterally and vertically relative to the motorcycle main frame and those movements are controlled by information from the **Control: Riding Position vs. Station (Open Loop)** or **Control: Riding Position vs. Time (Open Loop)** screen.

Overall, more details about the riding position control (both upper and lower body) are described in *Rider Controls (Motorcycle)*.

User Settings

① Rider upper body and lower body mass (keywords = `M_RIDER_BODY, M_RIDER_LEGS`). The rider's upper body includes torso, head arms, etc. The lower body includes legs and feet. The upper and lower body sections are divided approximately at the rider's hips.

② Moments and product of inertia of the rider's upper body. The moments and product are all taken about the mass center of the rider's upper body and are based on the rider upper body coordinate system in which the Z axis (vertical) is the rider body's spinal axis, Y axis (lateral)

is parallel with the ground when the vehicle is at rest on a flat level surface, and the X axis (longitudinal) is the cross product of Y and Z axis.

The moments of inertia in roll (`IXX_RIDER_BODY`), pitch (`IYY_RIDER_BODY`), and yaw (`IZZ_RIDER_BODY`) can be entered directly in these fields. Alternatively, they can be calculated using radii of gyration ④.

The XZ product of inertia about the mass center of the rider's upper body is defined as the negative of the volume integral:

$$I_{XZ} = -\int^{V} \rho \, x \, z \, dv$$

The product is positive when the principal X axis of the sprung mass tilts down (looking forward). This value must be entered directly—there are no built-in tools to calculate it.

③ Stiffness and damping associated with the rider's leaning (keywords = `K_RIDER_ROT`, `D_RIDER_ROT`). The rider may lean passively due to his mass and inertial forces. If so, the relative lean angle is determined using these parameters.

④ **Edit Radii of Gyration** checkbox. Checking this box enables editing of three data fields with X, Y, and Z radii of gyration ⑤, as an alternative to entering moments of inertia. When the box is checked, the moment of inertia fields are locked, but the values displayed in them are updated to reflect changes to the radii of gyration or mass. When it is unchecked, the moment of inertia fields are enabled for editing and the radii of gyration fields are locked, displaying values calculated from the mass and inertias.

⑤ Radii of gyration (not parameters). Sometimes measured values for `Ixx`, `Iyy`, and `Izz` are not available but `Rx`, `Ry`, and `Rz` can be estimated. When the **Edit Radii of Gyration** checkbox is checked, the corresponding moments of inertia are calculated with the equation:

$$I = M*R^2$$

where M is the rider's upper body mass ① and R is the specified radius of gyration.

Use any of these three radii if the corresponding moment of inertia is not available but the corresponding radius can be estimated.

⑥ Moments and product of inertia of the rider's lower body. The moments and product are all taken about the mass center of the rider's lower body and are based on a vehicle axis system in which the X axis (longitudinal) and Y axis (lateral) are parallel with the ground when the vehicle is at rest on a flat level surface, and the Z axis (vertical) points up in parallel with the gravity vector, and with the rider seated in his normal riding position.

The moments of inertia in roll (`IXX_RIDER_LEGS`), pitch (`IYY_RIDER_LEGS`), and yaw (`IZZ_RIDER_LEGS`) can be entered directly in these fields. Alternatively, they can be calculated using radii of gyration ⑧.

The XZ product of inertia about the mass center of the rider's lower body is defined as the negative of the volume integral:

$$I_{xz} = -\int^{V} \rho \, x \, z \, dv$$

The product is positive when the principal X axis of the sprung mass tilts down (looking forward). This value must be entered directly—there are no built-in tools to calculate it.

⑦ Stiffness and damping associated with the rider legs lateral movement (keywords = `K_RIDER_Y, D_RIDER_Y`). The rider may passively move in lateral relative to the main frame due to his mass and inertial forces. If so, the relative lateral displacement is determined using these parameters.

⑧ **Edit Radii of Gyration** checkbox. Checking this box enables editing of three data fields with X, Y, and Z radii of gyration ⑨, as an alternative to entering moments of inertia. When the box is checked, the moment of inertia fields are locked, but the values displayed in them are updated to reflect changes to the radii of gyration or mass. When it is unchecked, the moment of inertia fields are enabled for editing and the radii of gyration fields are locked, displaying values calculated from the mass and inertias

⑨ Radii of gyration (not parameters). Sometimes measured values for `Ixx`, `Iyy`, and `Izz` are not available but `Rx`, `Ry`, and `Rz` can be estimated. When the **Edit Radii of Gyration** checkbox is checked, the corresponding moments of inertia are calculated with the equation:

$$I = M*R^2$$

where M is the rider's lower body mass ① and R is the specified radius of gyration.

Use any of these three radii if the corresponding moment of inertia is not available but the corresponding radius can be estimated.

⑩ Distance that the rider body attachment (approximately the hip joint) lies behind the sprung mass origin (keyword = `LX_SEAT`). The orientation of the rider body mass is based on the location of this point.

⑪ Height of the rider body attachment (approximately the hip joint) above the origin (keyword = `H_SEAT`). The orientation of the rider body mass is based on the location of this point.

⑫ Distance that the rider's lower body mass center lies behind the sprung mass origin (keyword = `LX_CG_RIDER_LEGS`).

⑬ Height of the rider's lower body mass center above the origin (keyword = `H_CG_RIDER_LEGS`).

⑭ Height of the rider's upper body mass center above the rider body attachment (keyword = `H_CG_RIDER_BODY`) in the direction of the rider upper body coordinate system Z axis.

⑮ Angle that the rider leans forward from the motorcycle main frame vertical axis (keyword = `A_RIDER_BACK`). This is used to define the initial riding position and orientation of the upper body spinal axis.

⑯ Links to animator datasets. These are usually **Animator: Rider Shape Assembly**, but can be any dataset that defines the shape of the rider for the animator.

References

1. Sharp, R. S., Evangelou, S. and Limebeer, D. J. N., "Advances in the modelling of motorcycle dynamics", Multibody System Dynamics, 12(3), 2004, 251-283.

2. Sharp, R. S., Evangelou, S. and Limebeer, D. J. N., "Multibody aspects of motorcycle modelling with special reference to Autosim", Advances in Computational Multibody Systems, J. G. Ambrósio (Ed.), Springer-Verlag, Dordrecht, The Netherlands, 2005, 45-68.

3. Sayers, M.W., "Symbolic Vector/Dyadic Multibody Formalism for Tree-Topology Systems." Journal of Guidance, Control, and Dynamics, Vol. 14, No. 6, Nov/Dec 1991, 1240-1250.

4. Sayers, M.W., "Symbolic Computer Language for Multibody Systems." Journal of Guidance, Control, and Dynamics, Vol. 14, No. 6, Nov/Dec 1991, 1153-1163.